UNIVERSITY OF GRAZ
Department of Mathematics
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MAT.B0204UB MATHEMATICAL MODELLING
SUMMER SEMESTER 2025/26

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EXERCISE SHEET 4

Due: 13 May 2026, 10:00

EXERCISE 4.1
We consider the following second order ODE

$$mu'' + cu' + ku = 0, \tag{1}$$

with $u = u(t)$ and $c, k > 0$.

(i) Rewrite (1) in a 2 dimensional system, then perform the linear stability analysis of (1).

(ii) Compare the obtained result with the explicit solution.

EXERCISE 4.2
We consider the following system

$$\begin{cases} \dot{x} = -x - 2y + xy^2, \\ \dot{y} = 3x - 3y + y^3. \end{cases} \tag{2}$$

For the items (i) and (ii) below, determine the region where the origin is an asymptotically stable steady state of (2) and find the corresponding region of attraction.
Hint 1: Apply the Lyapunov theorem and use the Young inequality to estimate indefinite terms.
Hint 2: The region of attraction is bounded by the larger level set of the Lyapunov functional, satisfying the requested properties of the theorem.

(i) Consider the function $W(x, y) := \frac{1}{2}(x^2 + y^2)$.

(ii) Consider the function $W(x, y) := 3x^2 + 2y^2$.

(iii) Perform the linear stability analysis of (2) and compare the results with the Lyapunov method.

EXERCISE 4.3 *(Discrete Population Model: Cobwebs and Periodic Solutions)*
In this exercise, we study the behavior of the stationary solutions of the scaled discrete logistic growth as we vary the value of the system parameter r using a cobweb diagram. Thus, consider the discrete recursion given by

$$x_{n+1} = rx_n(1 - x_n), \qquad x(0) = x_0 \in (0, 1),$$

where $n \in \mathbb{N}$ and $r > 0$ is a proportionality parameter.

(i) Using the following values of x_0 and r, compute (e.g. by hand, MS Excel, Matlab, GeoGebra, Mathematica) the corresponding values of the sequence $\{x_n\}_{n\in\mathbb{N}}$ you obtain from the cobweb plot until a pattern is visible. Plot the points (n, x_n), $n = 0, 1, 2, ...$ on the Cartesian plane.

(a) $x_0 = 0.5, r = 0.5$

(b) $x_0 = 0.3, r = 1.5$

(c) $x_0 = 0.5, r = 3.2$

What is the behaviour of x_n as n increases?

(ii) Choose any starting point $x_0 \in (0,1)$ and graph (e.g. Matlab, GeoGebra, Mathematica) a cobweb diagram in the unit square $[0,1]^2$ for the following values of the parameter r. Plot the points (n, x_n), $n = 0, 1, 2, ...$ and discuss the behaviour of x_n as n increases.

(a) $0 < r < 1$

(b) $1 < r < 3$

(c) $3 < r < 1 + \sqrt{6} \approx 3.4494$. Try for $r = 3.1$ and $r = 3.4$ and you should be able to see periodic solutions (period = 2).

(d) $r > 1 + \sqrt{6} \approx 3.4494$. In this case, check whether the solution of the model has the following properties:

i. a cycle of period 4 for $r = 3.5$

ii. a cycle of period 8 for $r = 3.55$

iii. a cycle of period 16 for $r = 3.568$

iv. a cycle of period 3 for $r = 3.84$.

Explain what happens when $r = 3.6$ and when $r = 4$.